

ElectroRing 2.0: Precise Micro-Interaction Sensing for High-Bandwidth Wearable Input

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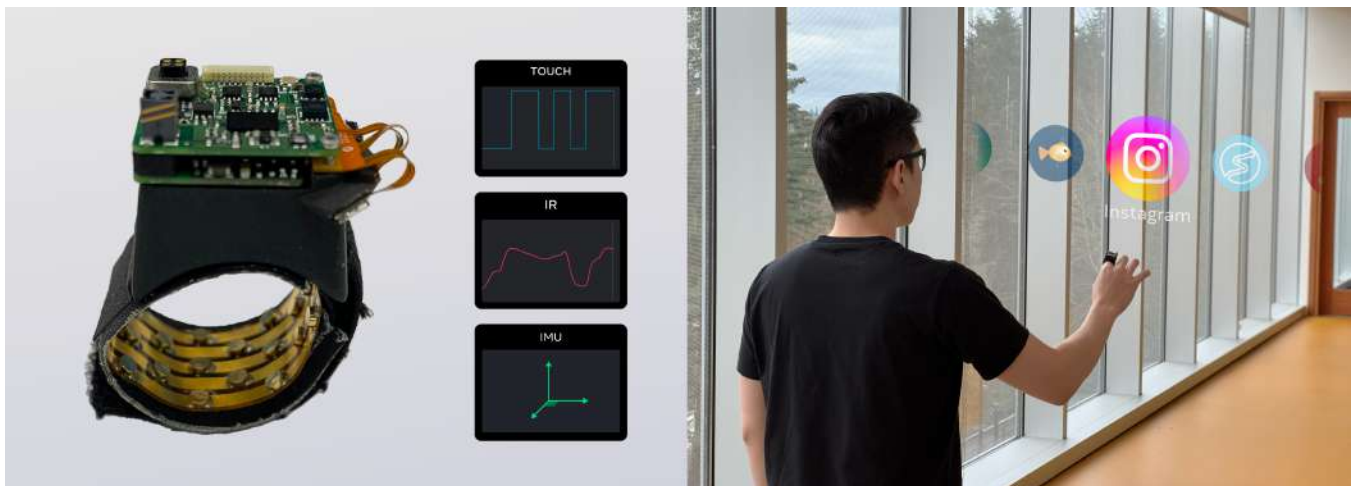


Figure 1: An overview of the ElectroRing 2.0 system. The left panel shows the finger-worn device with its integrated sensor hardware alongside real-time visualization of the multimodal sensor outputs, including touch (galvanic), infrared (IR), and inertial measurement unit (IMU) signals. The right panel shows a user wearing ElectroRing 2.0 and performing mid-air gestures to interact with a holographic augmented reality (AR) interface.

Abstract

We showcase a demo using ElectroRing 2.0, a multimodal, finger-worn input device for seamless, everyday AR/VR/AI interactions. Building on the fundamental sensing principle of ElectroRing [1], it integrates galvanic, inertial, and optical sensors to detect a rich set of micro-gestures, including scrolling, dial rotation, and more. ElectroRing 2.0 achieves over 95% precision and sub-40 ms latency, enabling robust, high-bandwidth input for wearable systems. We demonstrate ElectroRing 2.0's real-time performance and discuss its potential for next-generation AR/VR/AI interfaces. This is an

interactive demo in which users perform a variety of mid-air micro-gestures to control an AR interface in real time. The demo highlights ElectroRing 2.0's responsiveness, versatility, and ease of calibration, illustrating its potential for intuitive, high-bandwidth wearable input in practical scenarios.

CCS Concepts

• **Human-centered computing** → **Gestural input; Pointing; Mixed / augmented reality; Mobile devices.**

Keywords

Do, Not, Use, This, Code, Put, the, Correct, Terms, for, Your, Paper

ACM Reference Format:

Roger Boldu, Andre Levi, Eric Whitmire, Wolf Kienzle, and Hrvoje Benko. 2026. ElectroRing 2.0: Precise Micro-Interaction Sensing for High-Bandwidth Wearable Input. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona,



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CHI EA '26, Barcelona, Spain

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ACM ISBN 979-8-4007-2281-3/26/04
<https://doi.org/10.1145/3772363.3799171>

Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3799171>

1 Introduction

The rapid evolution of AR, VR, and AI glasses is driving a shift toward all-day wearable computing. Although advances in display and processing are important, the true success of these devices depends on how naturally and effectively users can interact with them. Users expect seamless, intuitive control that fits seamlessly into daily life, without relying on large, tiring gestures, voice commands, or cumbersome hardware. Current input solutions, especially those based on computer vision (CV) hand tracking, still face major limitations. CV systems often struggle with millimeter-level errors at the fingertip, making precise actions like scrolling or selecting small objects unreliable. More importantly, they cannot consistently detect the exact start and end of gestures, resulting in laggy or ambiguous interactions that undermine responsiveness and user experience. This gap between user expectations and current technology motivates our research question: How can we design a finger-worn input device that enables robust, high-bandwidth, and highly responsive interaction for all-day wearable AR/VR/AI systems?

In this work, we introduce ElectroRing 2.0, a state-of-the-art, multi-modal finger-worn input device that delivers unprecedented precision and reliability in gesture detection, especially in detecting the start and end boundaries of an interaction, so that interactions like scrolling and tapping become fluid and intuitive. Building on the foundational pinch detector presented in ElectroRing [1], this iteration uses a sensor-fusion approach to advance the field of mid-air gesture tracking by integrating galvanic, inertial, and external optical sensors into a small ring form factor. This allows it to capture a rich set of micro-gestures, including scrolling, pinch, rotation, thumb tap location, and compound movement. In this short paper, we highlight the sensing principles, engineering challenges, and system architecture that underpin ElectroRing 2.0. Our interactive demo showcases the system's real-time performance, demonstrating its potential to push the boundaries of ring-based input for AR/VR/AI.

The key contributions of this work are:

- **Design and implementation of ElectroRing 2.0:** A multi-modal finger-worn device that fuses galvanic, inertial, and optical data for comprehensive gesture capture.
- **Demonstration of rich, stateful micro-gesture detection:** Enables subtle and fluid interactions beyond binary touch, such as scrolling, rotation, and compound movements.
- **Empirical validation:** Evaluation of ElectroRing 2.0's performance in real-world scenarios, demonstrating its potential to enable high-bandwidth, all-day AR/VR/AI input.

2 Sensing Principle

2.1 Galvanic Coupling

ElectroRing [1] established the core principles of our approach. It utilized an active electrical (galvanic) sensing, in which a high-frequency AC signal (10 MHz) is coupled to the finger via a differential pair of electrodes inside the ring. When the user's fingertip touches their skin (e.g., in a thumb-to-index pinch), a circuit is

completed, causing a measurable, step-function-like change in the signal for both touch and release events.

This technique offers three primary advantages over other IMU-based methods or physical touch buttons:

- **High Signal-to-Noise Ratio (SNR):** The raw electrical signal exhibits a substantial SNR of approximately 25 dBV, making touch events clearly distinguishable from noise.
- **Subtlety of Detection:** It can reliably detect contact with minimal physical force or movement, a feat challenging for systems that rely on detecting the impact or visual change of a forceful tap.
- **Ergonomics:** Thumb-to-index finger mid-air interaction delivers higher ergonomics than IMU-based methods or physical button presses, as it requires minimal force and does not obstruct the fingertips.

ElectroRing's primary limitation was its inability to resolve finger state beyond binary contact. Although it could detect contact events with high precision, it was unable to determine the location of thumb contact or provide full six degrees of freedom (6DoF) tracking. ElectroRing 2.0 directly addresses this shortcoming through a multi-modal sensor fusion approach: it pairs an enhanced analog front end, achieving a higher SNR for more reliable touch detection, with an optical sensor for detecting thumb micro-gestures and a 6-DOF IMU for macro-motion context, together enabling a substantially richer vocabulary of interactions.

2.2 Multi-Modal Sensor Fusion

ElectroRing 2.0 not only detects when the fingertip and thumb are touching, but also fuses this information with optical and IMU data to determine where the finger is touching and what gestures the hand is performing—resulting in a much richer vocabulary of interactions. This fusion is designed to create an exceptionally rich and unambiguous data stream:

- The galvanic sensor provides a 200 Hz, temporally precise "snap line" for a touch event, establishing the exact moment of contact and release.
- A high-performance 6-DOF IMU provides macro-motion context, capturing whether a pinch was part of a larger dynamic gesture, like a wrist flick or 2D interaction.
- A miniature optical sensor captures subtle micro-motion from the thumb, including the location of the pinch on the finger or the trajectory of a slide.

3 Interactive Demo Description

For the demo, we introduce a series of interaction primitives and demonstrate how they can be used to navigate an AR interface. The accompanying video highlights how ElectroRing 2.0 serves as a prototyping tool, enabling seamless, low-friction end-to-end experiences. In this demonstration, ElectroRing 2.0 is used to control a smart glasses OS sandbox, showcasing ground-truth level performance with high-precision tracking and minimal latency.

These examples illustrate ElectroRing 2.0's versatility and responsiveness in supporting a wide range of AR interactions.

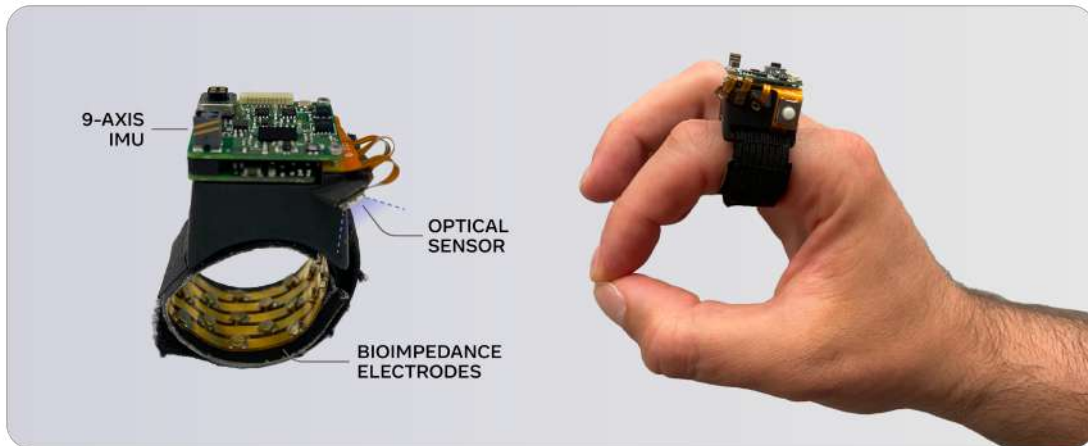
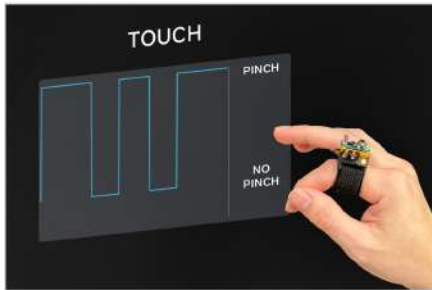


Figure 2: Hardware overview of ElectroRing 2.0, with the key sensing components labeled: the 9-axis IMU for motion capture, the optical sensor for thumb micro-motion tracking, and the bioimpedance electrodes for galvanic touch detection. Battery is not seen in the image, but it is 70mAh. The right image shows the assembled device worn on the index finger during a pinch gesture.

a) Touch (pinch)



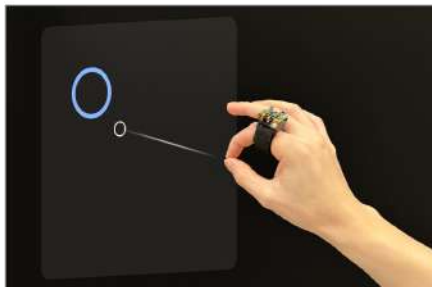
b) Thumb Position



c) Fluid Scrolling



d) 2D Pointing



e) 4-Way Wrist Flicks



f) Dial



Figure 3: A subset of interaction primitives supported by ElectroRing 2.0, including (a) Touch (pinch), (b) Thumb Position, (c) Fluid Scrolling, (d) 2D Pointing, (e) 4-Way Wrist Flicks, and (f) Dial, each shown with its corresponding real-time AR demo vignette.

3.1 Demo Setup and User Journey

The demo station consists of a host computer connected to ElectroRing 2.0 over BLE, with a large display for real-time visualization of sensor data and gesture recognition output. For certain scenarios, an XR headset may be used to demonstrate how these micro-gestures

can control applications in an immersive context. The user will proceed as follows:

- First, get fitted with ElectroRing 2.0 ring on user's index finger.

- Next, automated 15s calibration procedure, where the user performs 3-5 pinch and tap events.
- Finally, user is guided by on-screen prompts to perform a series of distinct micro-gestures, observing the system's live response.

4 Performance and User Experience

From a user experience standpoint, ElectroRing 2.0 interactions are highly fluid and responsive, with latency as low as 40 ms. Users consistently report high comfort and perceived accuracy, with high pinch gesture success rates and a short learning curve leading to strong gesture recall after several weeks of use. The unobtrusive design allows for natural interaction and minimal fatigue, while setup is straightforward, requiring Bluetooth pairing and calibration in under 15 seconds. Sensor fusion—combining bioimpedance, motion, and optical sensors—enhances reliability by reducing errors from movement and environmental changes.

By combining these modalities, ElectroRing 2.0 enables a spectrum of interactions—from simple pinches to complex, multi-modal gestures. The device's ability to sense and interpret subtle user actions, including the start and end of interactions, allows for rapid prototyping of next-generation interfaces. This delivers unprecedented responsiveness, resulting in high fluidity, control, and versatility in ElectroRing 2.0's capabilities.

Acknowledgments

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References

- [1] Wolf Kienzle, Eric Whitmire, Chris Rittaler, and Hrvoje Benko. 2021. ElectroRing: Subtle Pinch and Touch Detection with a Ring (*CHI '21*). New York, NY, USA, Article 3, 12 pages. doi:10.1145/3411764.3445094

Interaction Primitive	Description and Application Example
Touch	Detects pinch gestures (stateless or stateful) for selection, confirmation, and continuous control. Enables reliable, unobtrusive input without obstructing fingertips.
Thumb Position	Discrete selection between 3 touch locations (A/B stateful pinch), supporting multi-state input and contextual actions.
Thumb Swipes	Two discrete thumb-swipes (Left/Right) for quick navigation, menu traversal, or mode switching.
Dial / Knob Turn	Wrist roll combined with touch for rotational menu selection, volume adjustment, or scrolling through options.
2D Pointing	Wrist angle (IMU 6DoF) combined with touch for spatial pointing, cursor movement, or object manipulation in 2D space.
4-Way Wrist Flicks	Acceleration of wrist angle combined with on-off touch for rapid directional commands (e.g., up/down/left/right navigation).
Fluid Scrolling	Clutching plus wrist acceleration to smoothly navigate lists or content, supporting continuous and fluid scrolling.

Table 1: Subset of interaction primitives enabled by ElectroRing 2.0, illustrating their capabilities.